University of Nottingham
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School of Mathematical Sciences

Spring Semester 2025-2026

MATH1047 - MATHEMATICAL METHODS FOR CHEMICAL & ENVIRONMENTAL ENGINEERING

Coursework 4

Deadline: 6pm, Friday 15/5/2026

Your neat, clearly-legible solutions should be submitted electronically as a **IPYNB file** via the MATH1047 Moodle page by the deadline indicated there. A scan of a handwritten solution is acceptable. As this work is assessed, your submission must be entirely your own work (see the University's policy on Academic Misconduct); in particular, the University considers the unauthorised use of AI tools in assessments to constitute false authorship.

Submissions up to five working days late will be subject to a penalty of 5% of the maximum mark per working day.

Deadline extensions due to Support Plans and Extenuating Circumstances can be requested according to School and University policies, as applicable to this module.

Overview

This coursework assesses your understanding and implementation of two fundamental numerical algorithms:

1. Midpoint Newton Method – an iterative root-finding extension of Newton's method that achieves higher convergence order without requiring second derivatives.
2. Modified Midpoint Rule – a quadrature rule for approximating definite integrals.

You will complete the skeleton code provided in the Jupyter notebook **Coursework 4.ipynb** and use your implemented functions to compute specific numerical values.

Learning Objectives

By completing this coursework, you will:

- Translate multi-step mathematical formulas into working Python functions.
- Implement iterative methods with precise stopping criteria and loop control.
- Apply numerical integration rules that partition the integration domain.
- Use function arguments, loops, and return multiple values correctly.

Submission Requirements

- Please submit **a single IPYNB** (Jupyter Notebook) file. An IPYNB file to get you started is available to download from Moodle.
- The notebook should contain all code cells with your implementations.
- Include comments explaining key steps in your implementations.
- Name your file: ID_First name Surname .ipynb (e.g. 2022127_ Luyao Chen.ipynb). Please **double check the file name** before you submit it.
- This is an **individual coursework**, so you must write and submit your own work. Academic misconduct and late submissions will be penalized.

Marking Scheme:

Task 1: Midpoint Newton Method (25 marks).

- Correct implementation of the iteration formula (8 marks).
- Proper loop control (4 marks).
- Correct tolerance-based stopping (4 marks).
- Exact return format (4 marks).
- Code quality (clarity, meaningful comments, logical structure) (3 marks).
- Correct answer for **GR** (2 marks).

Task 2: Modified Midpoint Rule (25 marks)

- Correct computation of Δx and the nodes x_i (4 marks).
- Correct loop structure (6 marks).
- Correct evaluation of the summand $f(x_{3i-2}) + f(x_{3i-1})$ for each i (4 marks).
- Correct final multiplication by $\frac{3}{2}\Delta x$ and return of I (4 marks).
- Code Quality (clarity, meaningful comments, logical structure) (3 marks).
- Correct answer for **MID** (4 marks).